MIL-R-15472C(SHIPS)
23 March 1966
SUPERSEDING
MIL-R-15472B(SHIPS)
30 December 1958
(See 6.4)

MILITARY SPECIFICATION

RELAYS, MOTOR DRIVEN

1. SCOPE

1.1 Scope. - This specification covers motor driven relays for general alarm, chemical alarm and cease firing systems on Naval ships.

1.2 Classification. - The types of relays covered by this specification shall be designated in the following form (see 6.1).

IC/ - Basic indicator.
K - Type of operation.
1 - Function (see table I).
A - Type of current (see table II).

1.2.1 Basic indicator. - The basic indicator for Naval interior communication equipment is IC/.

1.2.2 Type of operation. - The equipment shall be motor operated. This is designated by the letter K.

1.2.3 Function. - The function is indicated by a number as shown in table I.

Table I - Function.

Designation	Function
1	General alarm and chemical alarm
2	Cease firing

1.2.4 Type of current. - The type of current is indicated by a letter as shown in table II.

Table II - Type of current.

Designation	Type of current.
A	115 volts alternating current (a.c.) - 60 cycles
B	115 volts a.c. - 400 cycles
C	115 volts direct current (d.c.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY
MIL-I-983 - Interior Communication Equipment, Naval Shipboard; Basic Design Requirements for.

FSC 5945

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MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

STANDARD

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply:

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd. St., New York 16, N.Y.)

3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production, one relay of each type shall be tested as specified in 4.2 (see 6.3).

3.2 General requirements. - Relays shall be in accordance with the following paragraphs of MIL-I-983 in addition to the requirements specified herein. When a requirement of MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern:

Materials, general
Substitution of (equal or superior) materials or parts
Fungus-inert materials
Unacceptable materials
Wood
Aluminum
Magnesium
Iron and steel
Plastics
Lubricants and lubrication
Painting
Bolts, machine screws, studs and nuts
Gaskets
Locking devices
Washers
Ball bearings
Parts, electrical - general
Capacitors
Variable resistors
Relays
Electrical tapes
Batteries
Enclosures - general
Enclosure - accessibility
Enclosure - degree of
Enclosure - mounting
Stiffening grooves
Through bolting
Cable entrance
Ventilation
Size
Threaded devices
Rounded corners and edges
Internal subassembly protection
Drilled and tapped holes
Welding
Temperature and humidity (maximum temperature rise does not apply to the actual contacts) (see 3.2.2)
Shock, vibration and inclination (for vital equipment)
Primary power supply circuits
Power supply tolerances
Personnel protection
Ground potential and grounding
Soldering
Internal subassembly connection
Terminal boards, connectors and terminals
Wiring
Dielectric strength and insulation resistance clearances
Airborne noise
Drawings, general
Interchangeability and standardization
Maintenance
Repair parts
Designation and marking
Workmanship and general examination

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3.2.1 The current-carrying springs shall be of beryllium copper or phosphor bronze in accordance with MIL-I-983.

3.2.2 The resistance between the primary contacts, when in the closed position, shall not exceed 0.01 ohm. The temperature rise of the contacts, at rated current and voltage, shall not exceed 50°C. over a 50°C. ambient for solid contacts or 65°C. over a 50°C. ambient for solid contacts, silver faced.

3.2.3 The relays shall be capable of passing the life test specified in 4.4.1.

3.2.4 Voltage and frequency tolerances. - The relays shall be designed to operate as required in 3.3.2 with voltage and frequency tolerances as follows:

Nominal voltage and frequency	Voltage tolerances	Frequency tolerances
115 a.c. 60 or 400 cycles	(See MIL-I-983)	(See MIL-I-983)
115 d.c. ----------	±7 volts	----------

3.3 Relay, motor-driven, for general alarm (type IC/K1A). - This relay shall consist of the necessary motor-driven cams, contacts and load relay mounted in a dripproof enclosing case. Enclosure shall be as specified in MIL-STD-108.

3.3.1 Operation. -

3.3.1.1 General alarm. - The closing of a manually operated general alarm switch at a remote location for a period of not more than 2 seconds shall energize the motor which shall, by means of the cams and contacts, cause the load relay to open and close the external load circuit at the rate of ninety times per minute (1/3 second closed and 1/3 second open) for a period of 15 seconds.

3.3.1.2 Chemical alarm. - The closing of a manually operated chemical alarm switch at a remote location shall energize the load relay and thus close the external load circuit contacts. The relay shall remain energized as long as the manually operated switch is held closed.

3.3.2 Features. - The external load circuit contacts shall be designed for continuous operation carrying a rated inductive load of 20 amperes at 115 volts a.c. and 0.5 factor. With the supply circuit energized but with the equipment in the nonoperating condition, there shall be no potential impressed on the motor. The equipment shall not become energized as a result of shock. Adequate provisions shall be incorporated in the design of the equipment to insure proper lubrication of the mechanism. When inclined as specified in 3.2 and with power supply tolerances as specified in 3.2.4, the relay shall not vary more than 15 percent from the time periods specified in 3.3.1.1.

3.3.3 Supply voltage. - Type IC/K1A relays shall be designed for operation on 115 volts, 60 cycles a.c.

3.4 Relay, motor-driven for general alarm (type IC/K1B). - The type IC/K1B relays shall comply with the requirements specified in 3.3, except that the relay shall be designed for operation on 115 volts, 400 cycles a.c.

3.5 Relay, motor-driven for general alarm (type IC/K1C). - The type IC/K1C relays shall comply with the requirements specified in 3.3 except that the external load circuits contacts shall be designed for continuous operation carrying a rated load of 20 amperes at 115 volts d.c., and the relay shall be designed for operation on 115 volts d.c. Two or more load circuits shall be provided. Each circuit shall be capable of carrying a proportionate amount of the rated load. A load relay shall be provided for each circuit.

3.6 Relay, motor-driven for cease firing (type IC/K2A). - This relay shall consist of the necessary motor-driven cams and contacts mounted in a dripproof enclosure in accordance with MIL-STD-108. This relay shall be identical to type IC/K1A, except for the following:

3.6.1 Operation. - The closing of a supply switch at the interior communication switchboard shall energize the motor which shall open and close the signal circuit contacts at a rate of ninety times a minute (1/3 second closed, 1/3 second open) as long as the supply switch is held closed.

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3.6.2 The external load circuit contacts shall be designed for continuous operation carrying a rated inductive load of 10 amperes at 115 volts a.c. and 0.8 power factor.

3.7 Relay, motor-driven for cease firing (type IC/K2B). - The type IC/K2B relays shall comply with the requirements specified in 3.6 except that the relays shall be designed for operation on 115 volts, 400 cycles a.c.

3.8 Repair parts kits. - Repair parts included in the repair parts kit shall be as follows:

2 sets of motor brushes.
1 motor armature (if wound type).
1 set of each type coil.
2 sets of electric contact.
1 of each type resistor, capacitor and spring.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection. - Preproduction inspection shall consist of the examination and tests listed in table III, which shall be performed in general in the order listed.

Table III - Preproduction inspection.

Test	Requirement paragraph	Test paragraph
General examination	3.2	4.4.2
Supply line voltage and frequency	3.2.4	MIL-I-983
Contact capacity	3.3.2, 3.4, 3.6.2	4.4.3
Contact temperature rise	3.2.2	4.4.4
Airborne noise	3.2	MIL-I-983
Temperature and humidity	3.2	MIL-I-983
Inclination	3.2	MIL-I-983
Dielectric strength	3.2	MIL-I-983
Insulation resistance	3.2	MIL-I-983
Life test	3.2.3	4.4.1
Enclosures	3.3 or 3.6	MIL-I-983
Vibration (vital)	3.2	MIL-I-983
Shock	3.2	MIL-I-983

4.3 Quality conformance inspection. - All relays shall be subjected to the quality conformance inspection listed in table IV, which shall be performed in general, in the order listed.

Table IV - Quality conformance inspection.

List	Requirement paragraph	Test paragraph
General examination	3.2	4.4.2
Supply line voltage and frequency	3.2.4	MIL-I-983
Enclosure	3.3 or 3.6	MIL-I-983

4.4 Test procedures. -

4.4.1 Life tests. - These tests shall be conducted as follows:

4.4.1.1 Type IC/K1A, IC/K1B and IC/K1C relays. - The relays shall be operated for a period of 30 hours at a rate of one operational cycle per minute (see 3.3.1.1) with the primary contacts carrying rated

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current (see 3.3.2 and 3.5). The entire life test shall be made at an ambient temperature of 130°F. The final hour of the life test shall be made with the primary contacts operating continuously at a rate of ninety impulses per minute carrying rated current. The temperature rise of the motor windings shall be measured during this part of the life test. Immediately following completion of the life run, contact resistance measurement shall be made to determine compliance with 3.2.2.

4.4.1.2 Type IC/K2A and IC/K2B relays. - The relay shall be operated for a period of 250 hours at continuous operation. The final hour of the life test shall be made with the primary contacts operating continuously at a rate of ninety impulses per minute carrying rated current (see 3.6.2). Primary contacts shall carry no load except during the final hour. Immediately following completion of the life run, contact resistance measurements shall be made to determine compliance with 3.2.2.

4.4.2 General examination. - The general examination shall be performed in accordance with Specification MIL-I-983. The operation test shall be in conformance with 3.3.1 or 3.6.1, as applicable.

4.4.3 Contact capacity test. - All relays shall be tested to determine that the current carrying capacity of the contacts comply with 3.3.2, 3.5 or 3.6.2 as applicable.

4.4.4 Contact temperature rise test. - All relay contacts shall be tested to determine compliance with the temperature rise specified in 3.2.2.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. -

5.1.1 Motor driven relays. -

5.1.1.1 Preservation and packaging. - Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. - The onboard repair parts shall be preserved and packaged level A; packed level C and marked levels A and C respectively in accordance with MIL-E-17555.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.2).

5.2.1.1 Preservation, packaging, packing and marking. - The equipment and repair parts shall be preserved and packaged levels A or C; packed level A or B as specified (see 6.2) and marked in accordance with MIL-E-17555.)

6. NOTES

6.1 Intended use. - Relays, motor-driven covered herein are currently used as follows:

Type	System	Supply voltage
IC/K1A	General alarm and chemical alarm	115 v., 1 phase, 60 cycle
IC/K1B	General alarm and chemical alarm	115 v., 1 phase, 400 cycles

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Type	System	Supply voltage
IC/K1C	General alarm and chemical alarm	115 v., direct current
IC/K2A	Cease firing	115 v., 1 phase, 60 cycle
IC/K2B	Cease firing	115 v., 1 phase, 400 cycles

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type number of the relays required (see 1.2).
(c) Preservation, packaging, packing or marking requirements other than those required by 5.1 (see 5.2).

6.3 Preproduction. -

6.3.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. - THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5945-N054SH)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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